Web Technologies Midterm Lab

Abdul Hadi | Registration number: **FA24-BCS-107**

N = 7 | BP = 607px | H = 49 degrees | K = 4

Task 1 - Mega-menu

Free delivery on orders over $50

THREAD STORE

WOMEN MEN ABOUT

HELP CART 0

▼ MENU

New in

All clothing

Knitwear

Denim

Collection

SHOP BY CATEGORY

▼ Women

- Coats
- Knitwear
- Dresses
- Tops

▶ Men

Mobile navigation and category accordions at 375px.

Free delivery on orders over $50

THREAD STORE

WOMEN MEN ABOUT

HELP CART 0

▶ MENU





Keyboard-focused four-column menu at 607px.

Free delivery on orders over $50

THREAD STORE

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▶ MENU





Keyboard-focused four-column menu at 1280px.

HTML added to the existing navigation

```
<div class="mega-item">

  <a class="mega-trigger" href="#categories">Shop by Category</a>

  <div class="mega-panel" id="categories">

    <section><h2>Women</h2><ul><li><a href="#products">Coats</a></li><li><a
href="#products">Knitwear</a></li><li><a
href="#products">Dresses</a></li><li><a
href="#products">Tops</a></li></ul></section>

    <section><h2>Men</h2><ul><li><a href="#products">Jackets</a></li><li><a
href="#products">Sweaters</a></li><li><a
href="#products">Shirts</a></li><li><a
href="#products">Trousers</a></li></ul></section>

    <section><h2>Accessories</h2><ul><li><a
href="#products">Bags</a></li><li><a href="#products">Belts</a></li><li><a
href="#products">Scarves</a></li><li><a
href="#products">Hats</a></li></ul></section>

    <section><h2>Collections</h2><ul><li><a href="#featured">New
arrivals</a></li><li><a href="#deals">Autumn edit</a></li><li><a
href="#products">Essentials</a></li><li><a
href="#products">Denim</a></li></ul></section>

  </div>

</div>



<nav class="mobile-menu" aria-label="Mobile shop navigation">

  <details>

    <summary>MENU</summary>

    <div class="mobile-menu-links">

      <a href="#featured">New in</a><a href="#products">All clothing</a><a
href="#products">Knitwear</a><a href="#products">Denim</a><a
href="#deals">Collection</a>

      <h2>Shop by Category</h2>

      <details><summary>Women</summary><ul><li><a
href="#products">Coats</a></li><li><a
href="#products">Knitwear</a></li><li><a
href="#products">Dresses</a></li><li><a
href="#products">Tops</a></li></ul></details>
```

```
      <details><summary>Men</summary><ul><li><a
href="#products">Jackets</a></li><li><a
href="#products">Sweaters</a></li><li><a
href="#products">Shirts</a></li><li><a
href="#products">Trousers</a></li></ul></details>

      <details><summary>Accessories</summary><ul><li><a
href="#products">Bags</a></li><li><a href="#products">Belts</a></li><li><a
href="#products">Scarves</a></li><li><a
href="#products">Hats</a></li></ul></details>

      <details><summary>Collections</summary><ul><li><a
href="#featured">New arrivals</a></li><li><a href="#deals">Autumn
edit</a></li><li><a href="#products">Essentials</a></li><li><a
href="#products">Denim</a></li></ul></details>

    </div>

  </details>

</nav>
```

CSS rules added

```
.menu { position: relative; z-index: 20; }

.menu-nav { display: flex; align-items: center; justify-content: center;
gap: 28px; font-size: 14px; }

.mega-item { position: static; }

.mega-trigger { display: inline-block; padding: 8px 0; }

.mega-panel { display: none; position: absolute; z-index: 1000; top: 100%;
left: 50%; width: 100vw; transform: translateX(-50%); grid-template-
columns: repeat(4, minmax(0, 1fr)); gap: 28px; padding: 28px max(24px,
calc((100vw - 1140px) / 2)); background: #fff; border-top: 1px solid #ddd;
box-shadow: 0 12px 18px rgba(0, 0, 0, .14); }

.mega-item:hover .mega-panel, .mega-item:focus-within .mega-panel {
display: grid; }

@media (max-width: 606px) { .menu { display: none; } .mobile-menu {
display: block; position: relative; z-index: 20; padding: 12px 20px;
background: #f5f5f5; border-bottom: 1px solid #e3e3e3; } .mobile-menu-links
{ display: grid; gap: 12px; padding: 16px 0 4px; } .mobile-menu details
details { padding: 8px 0; border-top: 1px solid #ddd; } }
```

Task 2 - Checkout form

Checkout

Contact

Full name

Email

Phone

© 2026 Thread Store. Abdul Hadi.

Stacked checkout layout at 375px.

Free delivery on orders over

WOMEN MEN ABOUT THREAD STO

Clothing / Checkout

Checkout

Contact

Full name

Email

Phone

Deliv

Address

City

Delivery

Continue

Two fieldsets side by side at 607px.

Free delivery on orders over

WOMEN MEN ABOUT THREAD STO

Clothing / Checkout

Checkout

Contact

Full name

Email

Phone

Deliv

Address

City

Delivery

Continue

Two fieldsets side by side at 1280px.

Free delivery on orde

WOMEN MEN ABOUT

THREAD S



Your order has b

Your order will be delivered or

Continue sho

Clothing About us Contact us

Language: English | Currency: USD

© 2026 Thread Store. Abdul Hadi.

Order confirmation with a green check and the selected delivery date.

checkout.html form markup

```
<form class="checkout-form" action="order-confirmation.html" method="get">
```

```
  <fieldset>
```

```
    <legend>Contact</legend>
```

```
    <div class="form-row"><label for="full-name">Full name</label><input
id="full-name" name="full-name" type="text" autocomplete="name"
placeholder=" " required /><small class="field-error">Enter your full
name.</small></div>
```

```
    <div class="form-row"><label for="email">Email</label><input id="email"
name="email" type="email" autocomplete="email" placeholder=" " required
/><small class="field-error">Enter a valid email address.</small></div>
```

```
    <div class="form-row"><label for="phone">Phone</label><input id="phone"
name="phone" type="tel" autocomplete="tel" pattern="03[0-9]{2}-[0-9]{7}"
placeholder=" " title="Use format 03XX-XXXXXXX" required /><small
class="field-error">Use phone format 03XX-XXXXXXX.</small></div>
```

```
  </fieldset>
```

```
  <fieldset>
```

```
    <legend>Delivery</legend>
```

```
    <div class="form-row"><label for="address">Address</label><input
id="address" name="address" type="text" autocomplete="street-address"
placeholder=" " required /><small class="field-error">Enter your delivery
address.</small></div>
```

```
    <div class="form-row"><label for="city">City</label><select id="city"
name="city" required><option value="" selected disabled>Select a
city</option><option>Lahore</option><option>Islamabad</option><option>Karac
hi</option><option>Faisalabad</option><option>Rawalpindi</option></select><
small class="field-error">Choose a delivery city.</small></div>
```

```
    <div class="form-row"><label for="delivery-date">Delivery
date</label><input id="delivery-date" name="delivery-date" type="date"
min="2026-10-29" max="2026-11-05" required /><small class="field-
error">Choose a date from October 29 to November 5, 2026.</small></div>
```

```
  </fieldset>
```

```
  <button class="btn shop-btn checkout-submit" type="submit">Place
Order</button>
```

```
</form>
```

checkout.js

```
const checkoutForm = document.querySelector(".checkout-form");
```

```
checkoutForm.addEventListener("submit", (event) => {
```

```
  event.preventDefault();
```

```
const deliveryDate = new FormData(checkoutForm).get("delivery-date");
```

```
  const confirmationUrl = new URL("order-confirmation.html",
window.location.href);
```

```
  confirmationUrl.searchParams.set("delivery-date", deliveryDate);
```

```
  localStorage.removeItem("threadStoreCart");
```

```
  window.location.assign(confirmationUrl);
```

```
});
```

order-confirmation.js

```
const dateValue = new
URLSearchParams(window.location.search).get("delivery-date");
```

```
const deliveryDate = document.querySelector("#delivery-date");
```

```
if (dateValue && /^\d{4}-\d{2}-\d{2}$/.test(dateValue)) {
```

```
  const parsedDate = new Date(`${dateValue}T00:00:00`);
```

```
  if (!Number.isNaN(parsedDate.getTime())) {
```

```
    deliveryDate.textContent = new Intl.DateTimeFormat("en-US", {
```

```
      year: "numeric",
```

```
      month: "long",
```

```
      day: "numeric",
```

```
    }).format(parsedDate);
```

```
    }
```

```
}
```

CSS rules added

```
.checkout-form { display: grid; grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 24px; }
```

```
.form-row { display: grid; grid-template-columns: 110px minmax(0, 1fr);
align-items: start; gap: 14px; margin: 12px 0; }
```

```
.form-row input:user-invalid, .form-row select:user-invalid { border-color:
#c62828; }
```

```
.field-error { display: none; grid-column: 2; color: #b71c1c; font-size:
13px; }
```

```
.form-row input:user-invalid + .field-error, .form-row select:user-invalid
+ .field-error { display: block; }
```

```
.checkout-submit { grid-column: 1 / -1; width: 100%; padding: 12px; }
```

```
@media (max-width: 606px) { .checkout-form { grid-template-columns: 1fr; }
.form-row { grid-template-columns: 1fr; gap: 4px; } .form-row label {
padding-top: 0; } .field-error { grid-column: 1; } }
```

```
@media (min-width: 607px) { .form-row { grid-template-columns: 110px
minmax(0, 1fr); } }
```

Task 3 - JavaScript cart

Fr

WOMEN MEN ABOUT TH

New in All clothing





Mini-cart with products and total at 375px.

Free delivery on orders over

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New in All clothing Knitwear Den



Your ca

Classic Be

Essential

Total



Mini-cart with products and total at 607px.

Free delivery on orders over

WOMEN MEN ABOUT **THREAD STO**

New in All clothing Knitwear Den

Your ca

Classic Be

Essential

Total





Mini-cart with products and total at 1280px.

script.js

```
const cartItems = [];
```

```
const productLimit = 5;
```

```
let nextCartItemId = 1;
```

```
const cartLink = document.querySelector(".cart-link");
```

```
const miniCart = document.querySelector("#mini-cart");
```

```
const miniCartItems = document.querySelector(".mini-cart-items");
```

```
const emptyMessage = document.querySelector(".mini-cart-empty");
```

```
const cartCount = document.querySelector(".cart-count");
```

```
const cartTotal = document.querySelector("#cart-total");
```

```
function renderCart() {
```

```
  miniCartItems.replaceChildren();
```

```
  cartItems.forEach((item) => {
```

```
    const row = document.createElement("li");
```

```
    row.className = "mini-cart-item";
```

```
    const name = document.createElement("span");
```

```
    name.textContent = item.name;
```

```
    const price = document.createElement("strong");
```

```
    price.textContent = `$${item.price.toFixed(2)}`;
```

```
    const removeButton = document.createElement("button");
```

```
    removeButton.type = "button";
```

```
    removeButton.dataset.itemId = item.id;
```

```
    removeButton.textContent = "Remove";
```

```
    row.append(name, price, removeButton);
```

```
    miniCartItems.append(row);
```

```
});
```

```
const total = cartItems.reduce((sum, item) => sum + item.price, 0);
```

```
  cartCount.textContent = cartItems.length;
```

```
  cartTotal.textContent = `$${total.toFixed(2)}`;
```

```
  emptyMessage.hidden = cartItems.length > 0;
```

```
  document.querySelectorAll(".item").forEach((card) => {
```

```
    const productName = card.querySelector("h3").textContent.trim();
```

```
    const addButton = card.querySelector(".shop-btn");
```

```
    const quantity = cartItems.filter((item) => item.name ===
productName).length;
```

```
    addButton.disabled = quantity >= productLimit;
```

```
    addButton.textContent = addButton.disabled ? "Limit reached" : "Add to
Cart";
```

```
});
```

```
}
```

```
document.querySelectorAll(".item").forEach((card) => {
```

```
  const addButton = card.querySelector(".shop-btn");
```

```
  addButton.addEventListener("click", () => {
```

```
    const name = card.querySelector("h3").textContent.trim();
```

```
    const priceText = card.querySelector(".price").textContent;
```

```
    const price = Number(priceText.replace(/[^0-9.]/g, ""));
```

```
    const quantity = cartItems.filter((item) => item.name === name).length;
```

```
    if (quantity < productLimit) {
```

```
    cartItems.push({ id: nextCartItemId, name, price });
```

```
    nextCartItemId += 1;
```

```
    renderCart();
```

```
    }
```

```
});
```

```
});
```

```
cartLink.addEventListener("click", (event) => {
```

```
  event.preventDefault();
```

```
  miniCart.hidden = !miniCart.hidden;
```

```
  cartLink.setAttribute("aria-expanded", String(!miniCart.hidden));
```

```
});
```

```
miniCartItems.addEventListener("click", (event) => {
```

```
  const removeButton = event.target.closest("button[data-item-id]");
```

```
  if (removeButton) {
```

```
    const itemIndex = cartItems.findIndex((item) => item.id ===
Number(removeButton.dataset.itemId));
```

```
    if (itemIndex !== -1) {
```

```
      cartItems.splice(itemIndex, 1);
```

```
    renderCart();
```

```
    }
```

```
    }
```

```
});
```

```
renderCart();
```

Mini-cart markup and CSS

```
<a class="cart cart-link" href="checkout.html" aria-expanded="false" aria-
controls="mini-cart">CART <span class="cart-count">0</span></a>
```

```
<section class="mini-cart" id="mini-cart" aria-label="Shopping cart"
hidden>
```

```
<h2>Your cart</h2><p class="mini-cart-empty">Your cart is empty.</p>
```

```
  <ul class="mini-cart-items"></ul>
```

```
  <p class="mini-cart-total">Total <strong id="cart-
total">$0.00</strong></p>
```

```
  <a class="btn shop-btn mini-cart-checkout" href="checkout.html">Go to
checkout</a>
```

```
</section>
```

```
.mini-cart { position: absolute; z-index: 1100; top: calc(100% + 14px);
right: 0; width: min(340px, calc(100vw - 32px)); padding: 18px; border: 1px
solid #ccc; background: #fff; box-shadow: 0 10px 24px rgba(0, 0, 0, .18); }
```

```
.mini-cart[hidden] { display: none; }
```

```
.mini-cart-items { display: grid; gap: 12px; margin: 0; padding: 0; list-style: none; }
```

```
.mini-cart-item { display: grid; grid-template-columns: minmax(0, 1fr)
auto; gap: 4px 12px; align-items: center; }
```

```
.mini-cart-total { display: flex; justify-content: space-between; }
```